

DUALITY



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SYNTHIFY X THE CENTRIFUGE



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OUR THEME - DUALITY



Duality is defined by the Oxford dictionary as ‘an instance of opposition or contrast between two concepts or two aspects of something’. This dicotomy is and has been present throughout our history and has fascinated societies ever since they formed. We find common themes like light and darkness; good and evil; earthily and etherial in virtually every culture.

These constructs arise from the direct observation of seemingly directly opposite phenomena: the cycles of day and night, mathematical duality in operations or , more recently, the wave-particle duality. All of these show that duality is more than a mere human-made concept: it is a recurring mechanisms that is present across dozens of knowledge fields. But why?

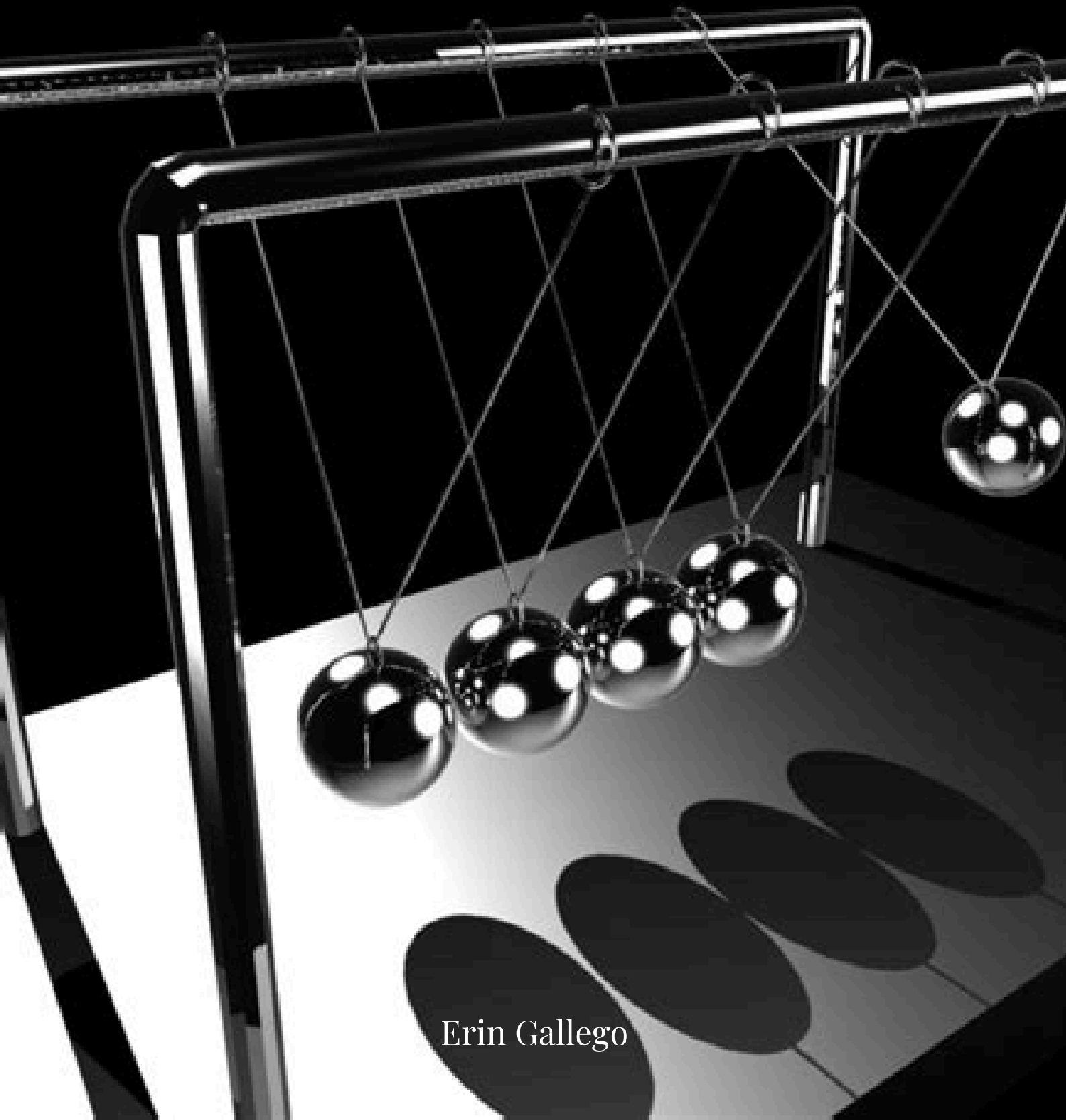
Duality beats the limits of a single-perspective framework of our world: by using a lens of dicotomy, it enables us to look past a single description, it creates links between concepts and makes them complimentary rather than contradictory. In doing so, it makes us be just a bit closer to the true richness of the universe.

Here we explore some of the roles this concept plays, helping us understand how our world works and the intricacies it entails - from the relationship between philosophy and maths, through psychossomatic open questions and the dicotomy between classical and quantum physics - we will explore some of the biggest contrasts in science an try to make sense of the duality that reigns our world.





THE MEASUREMENT PROBLEM: THE DUALITY BETWEEN PHYSICS AND PHILOSOPHY.



Erin Gallego

Introduction

For centuries, physics sought to explain the universe through objective laws, assuming systems have definite properties whether we look at them **or not**. But quantum mechanics, and its measurement problem, challenged that picture. By exploring key developments like Bell's theorem, Wheeler's thought experiments, and decoherence, this article examines how physics and philosophy intersect. **Ultimately, the measurement problem** remains unresolved scientifically, yet its philosophical impact is profound. Whether through Copenhagen, Many-Worlds, or QBism, interpretations reflect diverse ways of understanding reality **without changing physical predictions**. What it shows is that answering fundamental questions about the universe demands both empirical science and philosophical reasoning working together.

The Classical Belief

In classical physics, scientists operated under the philosophy of scientific realism. Scientific realism is the interpretation that the universe described by science exists independent of perception. Where objects possess definite properties, such as position and velocity, regardless of whether they are observed. An objective reality is the older belief that systems exist independent to human awareness, was therefore pivotal to classical physics.

Discovering Quantum

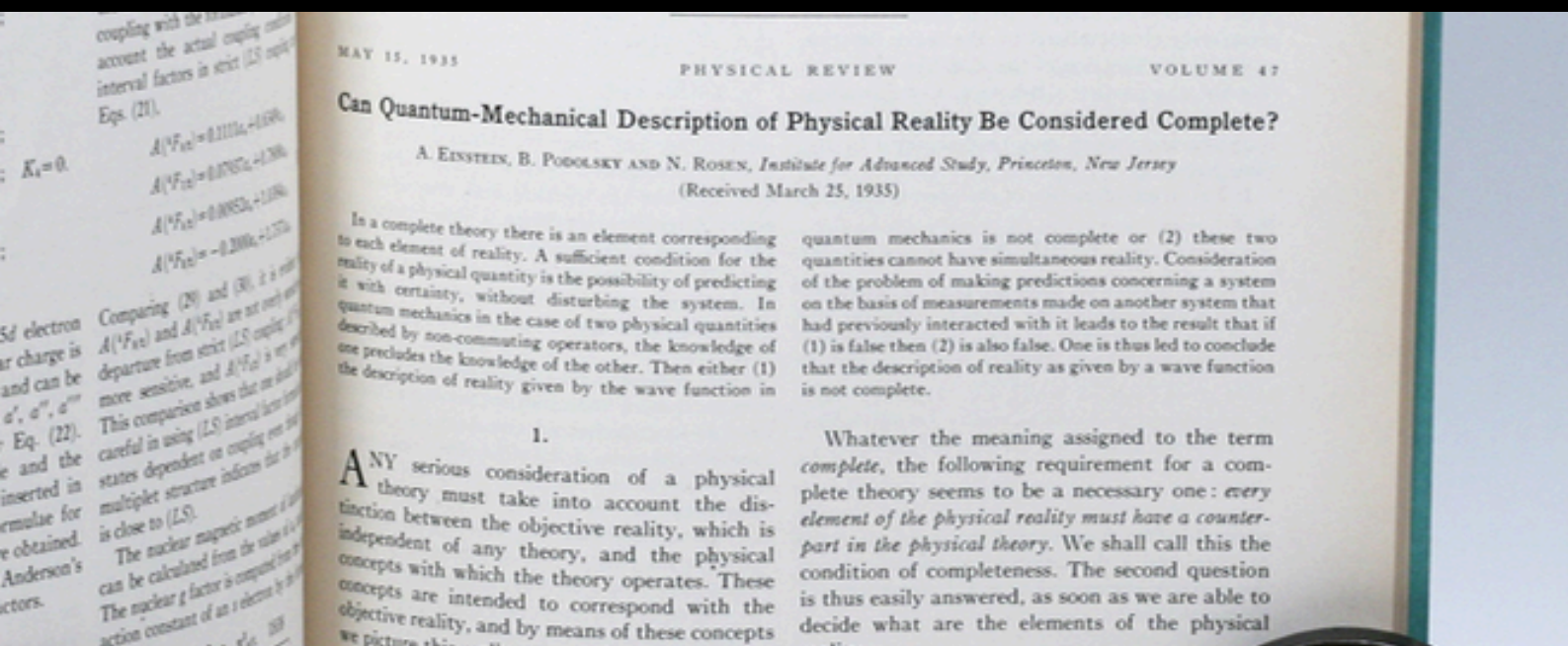
The founding of Planck's Law in 1900 was the first quantum theory in physics. Scientists quickly realised quantum mechanics was not compatible with realism. However, they were not prepared to abandon quantum mechanics as it was the best empirically confirmed scientific theory. There was a resistance to giving up realism in laboratory work, as they had to accept that the act of looking at subatomic particles alters them. Quantum mechanics does not describe things as they really are, since in the theory there are no 'things' in an absolute sense. This is explained when scientists discovered that particles do not occupy fixed positions in space; instead, the wave function describes the probability of finding said particle in a specific place. Before a measurement is made, a particle does not have a definite position or state; the wave function allows it to exist as a combination of multiple overlapping possibilities of its position/state at the same time. So, on a fundamental level, nature runs on probabilities, and therefore the classical view of physics cannot hold as systems do not have definite, observer-independent properties.

The Measurement Problem

The measurement problem explicitly relates to this crisis between classical physics and quantum mechanics. When a quantum system is observed or measured, it has a single, definite outcome, but prior to observation it existed as a mixture of multiple possible states (superposition). Quantum superposition is when a system can have this characteristic of existing in multiple states, behaving like a wave of distinct probabilities. Mathematically, quantum mechanics can predict the odds of where a particle may appear incredibly accurately, but it cannot explain the physical mechanism of this sudden transition.

The 1935 Paper (EPR)

Albert Einstein rejected the standard view of the measurement problem defined by the Copenhagen interpretation (explored in a later section). He did not believe that an observer's measurement caused a physical wave function to collapse, nor did he accept that subatomic particles lacked definite properties until they were observed. Instead, he argued that quantum mechanics was incomplete and that particles possess real, objective properties independently of any measurement. To defend the philosophy of scientific realism, he co-authored a paper in 1935 titled 'Can quantum-mechanical description of physical reality be considered complete'. He argued that if a particle displays a definite position upon observation, it must have that position all along. This was the first major philosophical revolt against the idea that measurement forces nature to choose a reality, showing how physics theories require philosophical dialogue to build logical explanations. This paper shows how quantum mechanics needs more than just maths to explain it, showing that philosophy is pivotal to understanding the universe.



Bell's Theorem

Einstein's critique persisted, but in 1964, John Bell devised a mathematical test to prove whether particles hold hidden properties before measurement. Quantum mechanics implies that measuring one entangled particle instantaneously fixes the state of another regardless of distance, creating an apparent conflict with special relativity's prohibition on faster-than-light communication. To resolve this paradox, Einstein, Podolsky, and Rosen (EPR) proposed that quantum mechanics was incomplete and that particles carry local "hidden variables" set at creation to dictate measurement outcomes without needing action at a distance, **as noted previously**. Niels Bohr countered that quantum mechanics was complete without such variables (his interpretation is known as the 'Copenhagen Interpretation' of quantum mechanics). In 1964, John Bell devised a mathematical test known as Bell's inequality, showing that violating this limit would rule out local hidden variable theories entirely. Specifically, Bell's theorem proves that no physical theory relying on local realism can reproduce every prediction of quantum mechanics. **Subsequent experiments consistently violated these inequalities**, proving that the universe cannot simultaneously be both local and deterministic. Subsequent experiments consistently violated Bell's inequalities, proving that the universe cannot simultaneously be local (no faster-than-light influences) and realistic (having definite properties before measurement). Regarding the measurement problem, this demonstrates that quantum correlations are not caused by hidden classical information, forcing us to accept that physical reality is either non-local or context-dependent on the act of measurement itself.

Wheeler's Delayed-Choice Experiment

John Wheeler took the implications of Bell's Theorem even further. His Delayed-Choice Experiment is a famous quantum mechanical thought experiment. It examines whether a photon 'decides' to act like a wave or a particle **based on how it is measured**, suggesting choosing a measurement type affects **past behaviour**. This interpretation links to the measurement problem as it deals with how a subatomic particle manifests itself after observation. But Wheeler also adds the thought of how the history of a particle is dictated by measuring it.

The von Neumann–Wigner Hypothesis

Another way to help fix the measurement problem was through the von Newman–Wigner hypothesis, which explores how human consciousness is strictly required to resolve a quantum superposition into a definite physical reality. This **hypothesis** is often summarised as ‘consciousness causes collapse.’ **John von Neumann showed mathematically that when a quantum system interacts with a measuring device, the tool becomes entangled with the system.** Physical laws treat the device and observer alike, **meaning** this chain of physical interaction never inherently stops by itself. Wigner argued the chain stops and collapses when the information registers in human subjective consciousness. Therefore, the mind is fundamental to quantum mechanics. Modern physicists reject this as the discovery of decoherence shows how systems lose wave interference by leaking information to the environment. There is also a philosophical critique that if consciousness makes reality, it is paradoxical how the universe existed billions of years before conscious life.

Could Decoherence Fix the Measurement Problem?

With respect to the measurement problem, decoherence provides an explanation for the transition of a system to an observable state. But decoherence does not claim to provide a mechanism for the actual wave-function collapse; instead, it puts forward a framework for the appearance of the wave-function collapse. If a system is not perfectly isolated, for example, during a measurement, coherence is shared with the environment and appears to be lost over time. Quantum coherence is not lost but mixed with many more degrees of freedom in the environment. **This is analogous to friction in classical mechanics, where energy is transferred as heat to the environment rather than disappearing entirely.** Although decoherence explains why superpositions become unobservable in everyday systems, it does not explain why we experience a single outcome, which gives both physics and philosophy a chance to keep working on this issue in quantum mechanics.

3 Philosophical Explanations

Quantum mechanics has left scientists unsure of a single theory about the nature of reality, **leading to several key interpretations:**

- The Copenhagen Interpretation is a leading theory brought forward mainly by Bohr and Heisenberg. **It attributes the indefinite state of a particle prior to observation to a fundamental limit on human knowledge.**
- Hugh Everett proposed the Many-Worlds Interpretation, saying that the wave function never collapses but instead evolves continuously everywhere. Every outcome that is possible for the quantum event does happen as the universe splits into multiple parallel, non-communicating realities. Regarding philosophy this interpretation changes ideas about the nature of probability and identity across branches of the universe.
- Quantum Bayesianism (QBism) merges Bayesian probability with quantum mechanics. Many aspects of quantum formalism are subjective in nature. For example, a quantum state is not an element of reality, representing degrees of belief an agent has about the possible outcomes of measurements. Philosophers of science deem this interpretation a form of anti-realism, the founders call QBism participatory realism which is when reality consists of more than can be captured by a third person account.

Conclusion

The measurement problem remains one of the greatest unresolved questions in modern science, sitting right at the crossroads of physics and philosophy. While experiments like those testing Bell's theorem rule out simple classical views of reality, they do not tell us which interpretation of quantum mechanics is correct. That is where philosophy **steps in to evaluate** what those physical facts really mean. In the end, the profound takeaway may be that neither discipline alone can give the full picture. The ongoing conversation between science and philosophy that edges us closer to understanding what reality really is.



Depression Mechanism

Body VS Brain or Body And Brain?



Mehreen Muzafar

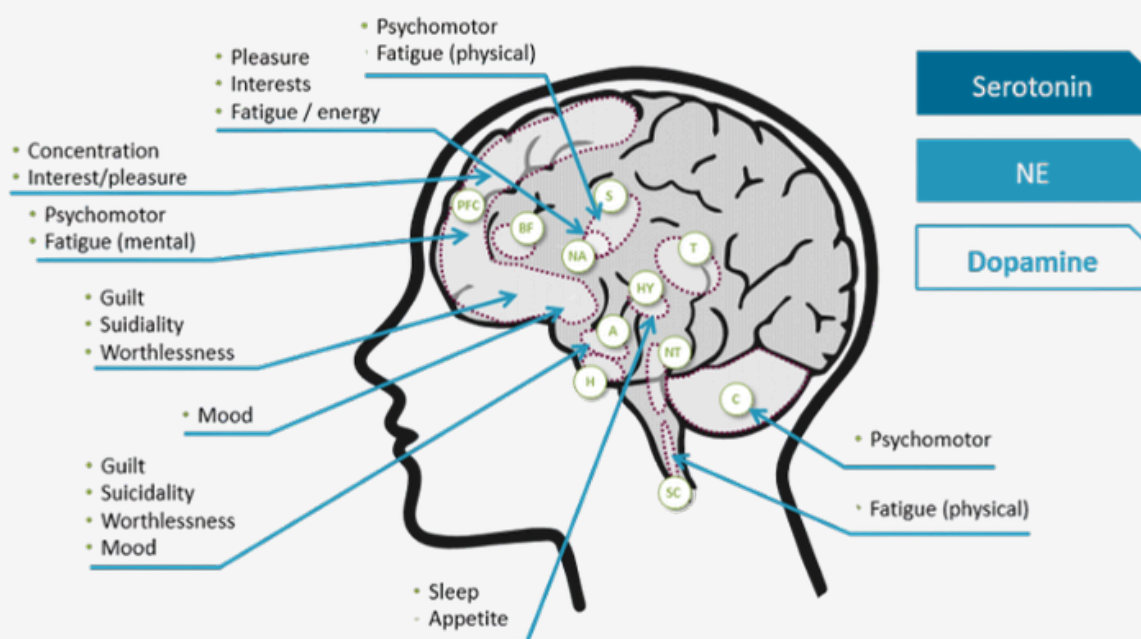
Introduction

Being told that the knot in your heart and the rot in your brain is just a chemical imbalance sounds pretty reassuring. It takes the blame off your soul and places it on your fragile body. You take some medicine, wait a few weeks, and get better.

Sounds pretty amazing, right? That's what millions of people were told for decades. But for almost thirty percent of them, this cure never worked.

In the early 1990s, a marketing campaign for Prozac promised a chemical cure for depression. The ads showed smiling people who had been restored to their old selves. It claimed that depression is a serotonin deficiency which can be easily fixed. But this claim, which came from a theory born in the 1960s, may have been only half the story.

Today, a new theory is emerging—one that links depression to an overactive immune system. And it's forcing us to rethink everything we were ever taught about depression mechanics.



The Monoamine Theory (Gen X Perspective)

The monoamine theory emerged in the 1960s after researchers noticed that certain drugs, like iproniazid and imipramine, seemed to improve mood by boosting levels of serotonin, norepinephrine, and dopamine. These drugs, which became the first antidepressants, worked by preventing the brain from breaking down or reabsorbing these chemicals.



For decades, this theory shaped everything about depression treatment. Pharmaceutical companies invested billions in developing SSRIs (Selective Serotonin Reuptake Inhibitors), which selectively target serotonin. The National Institute of Mental Health funded research that reinforced this theory, which ended up being widely accepted by the public. By the 1990s, Prozac had become a cultural phenomenon, prescribed to millions and featured on magazine covers.

But the monoamine theory has some serious gaps—SSRIs take two to four weeks to show any effect, even though they raise serotonin levels within hours. And about thirty percent of patients do not seem to respond to them at all.

So, the monoamine theory of depression, which dominated psychiatry for over fifty years, gave us SSRIs like Prozac and Zoloft, which have helped millions. But it also left millions without relief, unable to find a medication that worked for them.



The Macrophage Theory (Gen Z Perspective)

The macrophage theory offers a new way to look at depression by suggesting that long-term body inflammation triggers depressive symptoms.

It starts when the body's immune cells, called macrophages, release inflammatory proteins called cytokines. These cytokines travel through the body, cross the blood-brain barrier, and enter the brain. Once inside, specialized brain immune cells called microglia act as resident macrophages; when activated, they release specific cytokines like IL-6, TNF- α , and IL-1 β , causing brain inflammation that disrupts how the brain produces neurotransmitters.

This biological process explains why people with chronic inflammatory conditions like rheumatoid arthritis and Crohn's disease have higher rates of depression. This theory is backed by studies showing that people with depression have higher levels of these inflammatory markers in their blood, which explains why anti-inflammatory drugs show promise in treating depression and sometimes work faster than standard SSRIs..

What Both Theories Get Right and Wrong

The monoamine theory wasn't wrong in its approach when it gave us safe, effective medications that have helped millions. But it oversimplified depression into a chemical imbalance that could be fixed with one simple medication. This allowed pharmaceutical marketing to dominate the conversation and left little room for other explanations.

The macrophage theory offers hope for treatment-resistant patients and connects mental health to physical health, which feels intuitive to younger generations. But it is still emerging, and the evidence is correlational. It risks repeating the same pattern of reducing depression to one biological cause and selling a new drug to fix it.

The truth is very complicated, and both theories are likely true. Stress activates the immune system, which affects the brain. The immune response affects neurotransmitter production. And both can influence each other in a hard-to-break cycle.



**MIRACLE MOLECULE:
WHY WATER'S IMPORTANCE LIES
BEYOND NEUTRALITY**

Holly Curtaine

Introduction

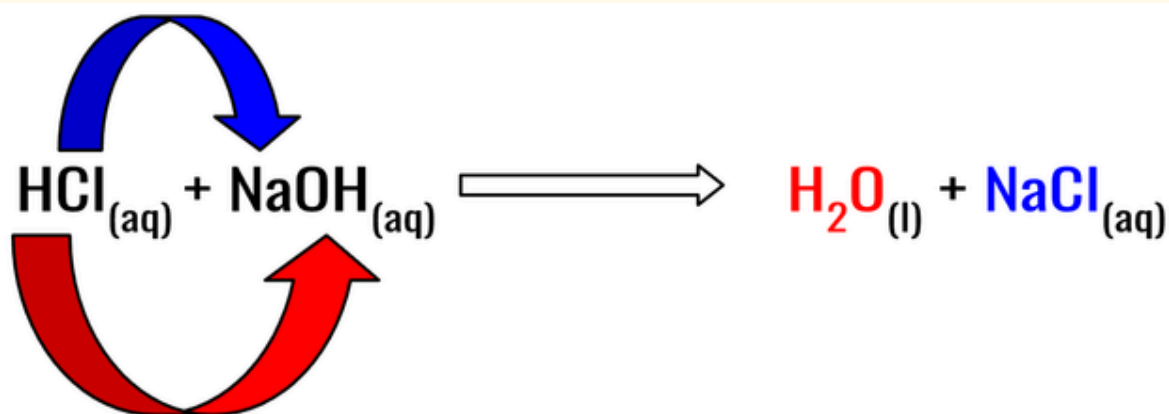
If you were asked to name the most abundant acid on Earth, what would you think of? Citric acid might come to mind - its presence in many citrus fruits as well as in cleaning products and even pharmaceuticals might make up a large volume worldwide. Another suggestion might be lactic acid, the acid that every human being produces when their cells break down glucose for energy. Surely the cells in all 8.3 billion of us produce enough acids to dominate everything else?

What if we flipped the question, asking instead what the most abundant base on Earth is? Ammonia might be a common answer due to its significance in agricultural fertilizers, cleaning products and manufacturing processes. Substances such as calcium hydroxide and sodium bicarbonate (known commonly as baking soda) are bases that appear in toothpaste, helping to protect tooth enamel from decay by neutralising acids produced by bacteria.

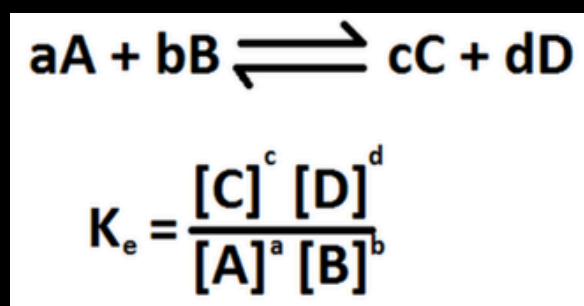
Surprisingly, the answer to these two questions is the same. Even more surprisingly, the answer is a substance known universally for its neutrality : water.

An Introduction to Acid-Base Chemistry

According to Bronsted-Lowry definitions, an acid is a substance that donates a proton (also known as a H^+ ion) to another species during a reaction. Conversely, a base is a substance that accepts a proton from another species. We can see this process more clearly when looking at the example of the reaction between hydrochloric acid (HCl) and sodium hydroxide (OH^-). HCl donates a H^+ ion to NaOH, with the OH^- ion accepting this proton to form water, leaving Na^+ and Cl^- to form sodium chloride.



There are also some mathematical concepts required for understanding water's significance. For a reversible reaction at equilibrium, the equilibrium constant is used to show the ratio between the concentrations of reactants and products at equilibrium, featuring a fraction with products on top and reactants on the bottom. The concentration of each substance is represented by square brackets, and each species is raised to the power of the coefficient in front of it in the balanced equation, as shown below. One important rule to note is that pure liquids and solids are not included in equilibrium calculations.



Finally, to find the pH of a substance, we use the formula $\text{pH} = -\log [\text{H}^+]$, where the logarithm is base 10. This is because the pH is a logarithmic scale, increasing by 1 every time the concentration of protons decreases by a power of 10.

Following this introduction to acids and bases, it soon becomes clear how water fits in with these ideas. It has 2 hydrogens that it could donate to another species, but also has 2 lone pairs of electrons, each ready to accept another proton. This results in its ability to act as an acid, for example when it reacts with ammonia by donating a proton to it, as well as its ability to be a base, for example when it accepts a proton during its reaction with hydrochloric acid. This dual ability is known as amphiproticity, with only a small number of substances having this same property.

Autoionization

As well as being able to react with other species as an acid or base, water can also react with itself in this way, in a process known as autoionisation. This lays the foundations for the pH scale and understanding it, making it significant in not only pH-sensitive biological processes, such as the regulation of blood pH in the human body, but also industrial applications such as water treatment.

The autoionisation of water occurs when one water molecule, acting as an acid, donates a proton to another water molecule, acting as a base. This leads to the formation of a H_3O^+ ion and an OH^- ion, which react to form water again, leading to the establishment of an equilibrium with the equation $\text{H}_2\text{O}(\text{l}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{H}_3\text{O}^+(\text{aq}) + \text{OH}^-(\text{aq})$. We can write an expression for the equilibrium constant, ignoring the concentration of water as it is in the liquid phase, as $K_c = [\text{H}_3\text{O}^+][\text{OH}^-]$. This is more commonly denoted as K_w , and has a value of exactly $1.00 \times 10^{-14} \text{ mol}^2\text{dm}^{-6}$ at 25°C .

This concept helps us to understand why pure water has a pH of exactly 7. In pure water, $[\text{H}^+] = [\text{OH}^-]$, and so assigning this value as x results in the expression $x^2 = 1.00 \times 10^{-14}$, leading us to find that $x = 1.00 \times 10^{-7}$. Taking x to equal $[\text{H}^+]$, we can find pH by doing $\text{pH} = -\log[\text{H}^+]$, giving $-\log(1.00 \times 10^{-7}) = 7$. This relationship forms the basis of the pH scale, placing water exactly in the centre between 0 and 14, giving scientists a method for measuring the alkalinity and acidity of substances. The applications of this pH scale are incredibly valuable to a range of industries, in particular when it comes to pharmaceuticals. When developing drugs, pharmacologists need to be aware of the various pH levels in the body and how these can affect drug performance. For example, the pH of the stomach is around 1-2 while the pH of the small intestine lies between 5 and 7, meaning that weakly basic compounds dissolve more easily in the stomach while weakly acidic compounds dissolve more easily in the intestines. This has led scientists to alter the conditions under which certain oral drugs are taken to influence their absorption. An example is the coadministration of Posaconazole, a slightly basic compound taken as a treatment for invasive fungal infections, with a cola beverage of pH 2.5, as the lower pH of the cola increases the solubility of the drug in the stomach, boosting its absorption.

Another important application of pH in pharmaceuticals is understanding protein crystallization, as various pH levels can affect the structure and solubility of proteins. Investigations focused on human insulin found that higher pH levels increased its solubility, an important property that ensures it can move into the blood quickly to manage blood sugar levels. This holds significance when it comes to the use of insulin as a treatment for diabetics, as while short-acting insulin must be soluble so it can be taken quickly before meals to prevent blood glucose levels being too high when carbohydrates are eaten, long-acting insulin should not be as soluble so it can work slowly to keep levels stable between meals, meaning it is important that pharmacologists understand the nuance between the pH values of different forms of insulin.

Conclusion

In conclusion, water's amphiproticity cannot be underestimated. From pharmaceuticals to agriculture, the ability of water to be an acid or base plays a crucial role in supporting the global economy and human life on Earth. It might remain as a symbol of neutrality, seen for its support for life on Earth, but as chemists of the future, we need to see water beyond this lens if we want to take full advantage of its properties.



As a result, it can clearly be seen that the foundations laid by the autoionisation of water for the pH scale have been crucial for pharmaceutical development, as without clearly defined parameters, it would make interpreting and altering acidity and alkalinity much more difficult in situations where precision is essential.

Acting as an acid

The importance of water's amphiproticity does not solely lie in its ability to react with itself, as an equally significant property is its ability to act as an acid or a base when reacting with other substances. One situation is water's reaction with ammonia (NH_3) to form ammonium hydroxide (NH_4OH), also known as ammonia solution. Water, acting as an acid, donates a proton to ammonia to form NH_4^+ and OH^- . This reaction lays the foundation for numerous agricultural and manufacturing processes. One such use of ammonia solution is its use as a cleaning product, with its unique property of being a gas dissolved in water meaning that when wiped on a surface, the ammonia evaporates along with the water, which results in the streak-free finish desired by so many household cleaners. However, the largest use of ammonia solution lies in agriculture, with around 80% of all ammonia produced going towards this purpose. The nitrogen in ammonia is essential for plant growth as it plays a key role in the production of chlorophyll, the green pigment that traps sunlight for photosynthesis, as well as being important for protein synthesis, facilitating growth and the development of structures such as leaves and seeds. Without ammonia's reaction with water, anhydrous ammonia would have to be used by farmers, which is a dangerous substance due to its need to be stored under high pressure as a liquid. This holds the risk of a large, toxic gas cloud if its container breaks, making it a much more risky substance to store and use, especially as this liquid pressure form must be injected deep into soil due to possibly violent reactions with water, minerals and other substances in the earth. In contrast, the use of ammonia solution makes the process easier and safer as it does not have to be injected as far into the soil, and can be stored at a much lower pressure. Subsequently, it can be clearly seen that water's reaction with ammonia plays a crucial role in the production of crops due to the ease of use and safety of ammonia solution, helping to ensure food supplies can grow to meet the global demand.



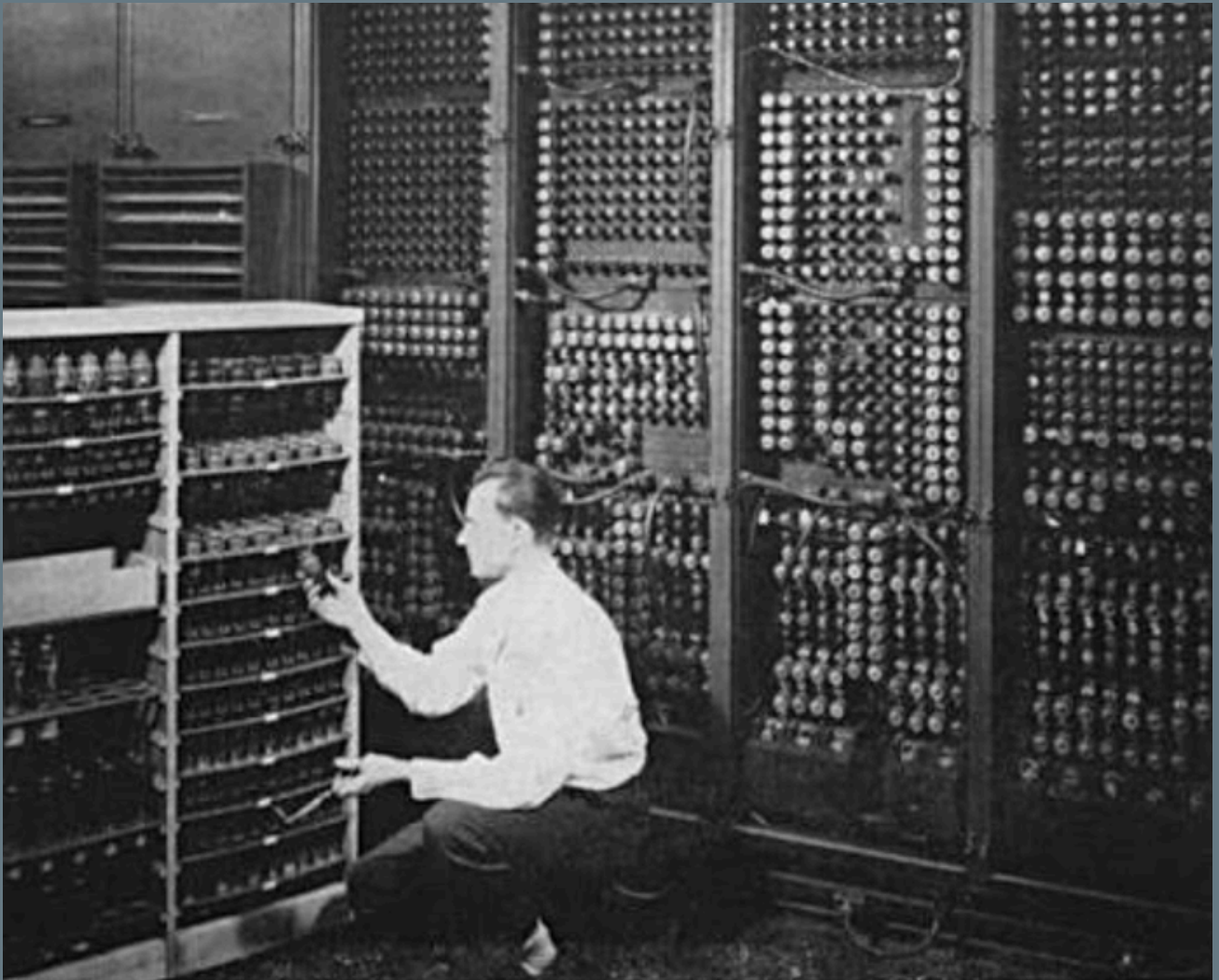
QUANTUM VS.



Classical Computing

Revolution or Evolution?

Serena Li



How often do you get mad at your computer for being too slow? Laggy webpages, sluggish file downloads and unresponsive buttons are common anguishes of classical computers that unleash the worst in all of us. Quantum computing is often marketed as the cornerstone of the next generation of computation, promising to solve in minutes tasks that would take even the most powerful classical supercomputers billions of years to solve. So then, could these gigantic refrigerator-sized machines be replacements, or supplements, for the computers of today?

To answer this question, it's important to first understand the fundamental mechanisms that govern how these computers work.

Classical Computing

Classical computing, which is what powers the desktops and laptops we are familiar with, involves executing actions by interpreting binary code. This consists of vast arrays of 1s and 0s, known as bits, which are the smallest representations of information in a computer. These two values correspond to two distinguishable physical states within electronic circuits, typically represented by high and low voltage. Using only two states makes information highly reliable to store, transmit and manipulate, since a circuit only needs to distinguish between two voltage ranges rather than many.



Millions or even billions of tiny components, known as transistors, control these electric signals. By performing logical operations on bits (such as AND, OR, NOT), these signals can be used to represent much more complex and useful information. Complicated operations, such as rendering a webpage, are constructed from the combination of enormous amounts of these simple binary operations.

So who carries out these operations?

Computers usually contain a chip called the central processing unit (CPU), which acts as the main “brain” of the device. It executes instructions and coordinates most of the computer’s operations. Alongside it, many computers use a graphics processing unit (GPU), which specialises in performing large numbers of calculations simultaneously. Although originally designed to render images, graphics and video, GPUs are now also used for computationally intensive tasks such as artificial intelligence and scientific simulations. Modern CPUs and GPUs can perform billions or even trillions of operations per second, which is more than sufficient for most everyday tasks such as browsing webpages, playing videos and running applications.

Sometimes, slow performance occurs when these components are overloaded with more computations than they can efficiently process, or when the computer has insufficient memory available. Yet despite their remarkable speed, classical computers face a more fundamental limitation: as certain problems grow in complexity, the number of calculations required can increase so dramatically that even the world's most powerful supercomputers become impractically slow. Could quantum computing be the solution?

Quantum Computing

Quantum computers are able to approach certain problems in fundamentally different ways by exploiting the physical laws that govern matter at extremely small scales. Rather than simply making calculations faster, they use quantum phenomena to represent and manipulate information in ways that classical computers cannot efficiently reproduce.



In quantum computing, information is stored in units called qubits, which parallel the function of bits in classical computing. However, instead of existing as either a 0 or a 1, a qubit can exist as a superposition of both values simultaneously. Furthermore, when multiple qubits interact, their states may also become “entangled”. This means that their quantum states become interconnected and must be described in relation to one another. But although it may sound like we can simply test every possibility simultaneously with the power of superposition in these qubits and immediately reveal the correct answer to a question, there is a catch: when the qubits are measured, their superposition collapses, leaving us with only one definite outcome. Quantum algorithms therefore manipulate the probabilities associated with these possible outcomes before measurement, using quantum interference to amplify useful results and suppress others.

Quantum states behave somewhat like waves, as they can reinforce or cancel one another through a process called quantum interference. You might think of this like an artist working on a drawing, strengthening certain lines while erasing others until the intended image becomes clearer. Similarly, quantum algorithms manipulate interfering states so that pathways leading towards useful answers are strengthened, while others are suppressed. When the qubits are finally measured, the desired result is therefore more likely to emerge. Quantum algorithms that feature this mathematical manipulation to reveal useful structures and hidden correlations are extremely powerful for certain tasks.

The first popular application of this technology is directly derived from the nature of quantum computing. Molecules, atoms and electrons already obey the laws of quantum mechanics, so attempting to simulate them using classical bits can become extraordinarily computationally demanding. A quantum computer instead operates according to those same underlying principles, potentially allowing scientists to model complex molecules, chemical reactions and new materials far more efficiently.

The second application is less obvious, but potentially just as significant. Certain quantum algorithms can exploit hidden mathematical structures within enormous sets of possibilities. Shor's algorithm, for example, can theoretically factor sufficiently large integers dramatically more efficiently than the best known classical algorithms, posing a threat to some widely used forms of encryption. So quantum computers are not simply extraordinarily fast versions of classical computers. Their advantage comes from representing and processing particular problems in an entirely different mathematical and physical framework.

These properties make quantum computers particularly promising for two broad types of tasks: modelling physical systems and identifying mathematical patterns and structures in information. Because nature itself behaves quantum mechanically at atomic and molecular scales, quantum computers may be particularly well suited to simulating molecules, materials and chemical reactions. Meanwhile, algorithms such as Shor's algorithm demonstrate how quantum effects can expose mathematical structures that are extremely difficult for classical computers to find.

Why Classical Computing Is Not Disappearing

With all the high-stakes excitement surrounding qubits and quantum supremacy, it is easy to fall into the trap of assuming classical computers are headed straight for the history books. However, the idea that a quantum refrigerator will one day replace your smartphone rests on a fundamental misconception: quantum computers are not simply "faster" classical computers. They are specialized tools built for very specific types of mathematical problems.

For the vast majority of tasks we rely on daily, such as writing documents, streaming high-definition video, running local operating systems, or rendering modern video games, classical architecture remains vastly superior. A CPU is designed to execute linear, highly complex sequences of logical instructions with incredible clock speeds and near-zero error rates. Attempting to load a web browser on a quantum computer would offer no performance benefit whatsoever; in fact, the overhead of translating those tasks into quantum circuits would make the experience agonizingly slow.

Furthermore, qubits are notoriously delicate physical entities. Unlike classical transistors, which reliably maintain their high or low voltage states over years of continuous operation, qubits are susceptible to a phenomenon called environmental decoherence. Tiny fluctuations in temperature, magnetic fields, or cosmic radiation can cause a qubit to lose its quantum state, introducing severe errors into a calculation.

To overcome this fragility, scientists rely on quantum error correction, a technique that pools hundreds or even thousands of delicate physical qubits together just to create a single, stable "logical qubit." Constructing a fault-tolerant quantum computer capable of running long, complex algorithms requires massive physical hardware, extreme cooling systems operating near absolute zero, and vast computational infrastructure. For everyday computation, classical bits are not just good enough...they are infinitely more practical, stable, and cost-effective.

Revolution or Evolution?

So, how should we characterize the rise of quantum technology? Is it a sudden revolution that upends the computing paradigm, or a natural evolution of our technological capabilities?

The answer is both. Quantum computing is undoubtedly a revolution in computational theory. By shifting from classical Boolean logic to the probabilistic mechanics of Hilbert space, computer scientists have unlocked entirely new complexity classes. Algorithms like Grover's and Shor's prove that certain problems previously deemed intractable in human lifetimes can now be solved in a matter of hours. This shifts our understanding of what is physically computable, completely revolutionizing fields like cryptography, quantum chemistry, and structural material design.

Yet, from an architectural standpoint, the integration of quantum computing represents an evolutionary step in hybrid computing. Rather than replacing the classical framework, quantum devices will almost certainly function as specialized coprocessors, much like how GPUs evolved alongside CPUs to handle heavy parallel graphics processing.

In a future enterprise setup, a classical supercomputer will likely manage the overall workflow, feeding specific, highly complex mathematical bottlenecks to a Quantum Processing Unit (QPU). Once the QPU completes its probabilistic operations and measures the collapse of its qubits, it will hand the finalized data back to the classical host. The two paradigms will not destroy one another; they will collaborate.

Conclusion

Quantum computing challenges the long-held boundaries of what classical machines can achieve, fundamentally reshaping our approach to optimization, cryptography, and physical simulation. Yet, far from rendering binary code obsolete, it highlights just how remarkably efficient and versatile classical computers truly are.

The future of technology will not be defined by a zero-sum battle between 0s and 1s and superpositions of both. Instead, the next era of innovation lies in synergy; a hybrid reality where classical systems handle the structural bedrock of daily processing, while quantum accelerators tackle the deep mysteries of the microscopic world. Ultimately, the question isn't whether the future belongs to quantum or classical computing, but rather how much further we can go when we harness the power of both.



DUPLICATES: EQUATIONS AND EXISTENCE.

Gianluca Grassi

Our world is littered with duplicates. Lots of what we consume on a daily basis is in fact a replica of something more simple, perhaps with added bells and whistles. For example, the pride of our generation, an iced shaken caramel latte frappe (the list of adjectives goes on) is just coffee and syrups for the most part. In physics and mathematics, many equations we thought to be singular are in fact a part of a family of equations that really mean the same thing, but can still be useful in their own right in different situations.

A prime example of this is one of the most influential and important equations in mechanical physics: Newton's Second Law of Motion. Many people are content with the idea that Newton's Second Law is:

$$F = ma$$

but few appreciate the finer details that brought us to this simplification. In reality, the law states:

$$F = \frac{dP}{dt}$$

This means that the force (F) is equal to the rate of change of momentum (P). As momentum is the same as mass multiplied by velocity, we obtain this equation:

$$F = \frac{d(mv)}{dt}$$

Using calculus, the derivative of a product of two functions (here m and v) can be written, using the product rule, as:

$$\frac{dm}{dt}(v) + \frac{dv}{dt}(m)$$

Examining the first term, we see that there is a rate of change of mass. In most systems, this will be equivalent to zero as the object will not lose mass as it moves. This assumption leaves us with

$$F = \frac{dv}{dt} \times m$$

The rate of change of velocity is also known as acceleration (a) which neatly brings us back to:

$$F = ma$$

However, note that this is only true when the rate of change of mass is zero and therefore will be false for an object with changing mass such as a rocket using fuel.

But what does this equation really mean? Known as the cause and effect equation- and helpfully rearranged to

$$a = \frac{F}{m}$$

it is very useful in explaining how objects move. Firstly when F, the resultant force on an object is zero then the acceleration must also be zero to balance the equation, an idea echoed by Newton's First Law. It also means that when an object has a larger mass, the acceleration will be lower for the same amount of force as mass is on the denominator. This resistance to acceleration characterised by a higher mass is called inertia.

With our knowledge of $F=ma$ and calculus, we can derive all kinematic equations from Newton's Second Law starting with the definition of acceleration:

$$a = \frac{dv}{dt}$$
$$adt = dv$$

After rearranging, integrating both sides with limits at the initial and final position gives:

$$\int_{t_i}^{t_f} a dt = \int_{v_i}^{v_f} dv$$
$$a[t_f - t_i] = v_f - v_i$$

Normally in kinematics we refer to the initial velocity, v_i , as “u” and the final velocity, v_f as v:

$$a \times \Delta t = v - u$$
$$v = u + at$$

At last we have derived the first of three kinematic equations, showing that while it may contain different terms, it is simply the same as the definition of acceleration, a tool for calculations of objects in motion governed by Newton's Second Law.

Using the equation we have just determined, we can use it and the definition of velocity as a springboard to find two other kinematic equations:

$$v = \frac{ds}{dt}$$
$$vdt = ds$$
$$v = u + at$$

Integrating both sides with respect to t allows us to substitute ds into the equation:

$$\begin{aligned}\int v dt &= \int (u + at) dt \\ \int_{s_i}^{s_f} v dt &= ut + \frac{1}{2}at^2 \\ \Delta s &= ut + \frac{1}{2}at^2\end{aligned}$$

Which brings us neatly to the second equation.

The third equation requires equating dt:

$$\begin{aligned}a &= \frac{dv}{dt} \\ dt &= \frac{dv}{a} \\ v &= \frac{ds}{dt} \\ dt &= \frac{ds}{v}\end{aligned}$$

Now that we have two different expressions for dt, we can set them as equal to each other:

$$\frac{ds}{v} = \frac{dv}{a}$$

Integrating both sides...

$$\int_{v_i}^{v_f} v dv = \int_{s_i}^{s_f} a ds$$

$$\left[\frac{v^2}{2} \right]_{v_i}^{v_f} = [as]_{s_i}^{s_f}$$

$$\frac{1}{2}(v_f^2 - v_i^2) = a(s_f - s_i)$$

Remembering that $v_f = v$ and $v_i = u$, and writing $(s_f - s_i)$ as Δs or just s ...

$$\frac{1}{2}(v^2 - u^2) = as$$

$$v^2 = u^2 + 2as$$

Our third and final equation!

Beyond being an enjoyable exercise in the fluidity of maths and the joys of figuring out a derivation like pieces coming together in a puzzle, this demonstrates the multifaceted nature of equations themselves in that they are often culminations or simplifications of other laws, rules or expressions. Often, we find many equations that mean the same thing but are useful in different situations such as these kinematic equations.

In conclusion, the world and, of course, the scientific laws that govern it are littered with duplicates. This however is not a bad thing, and I hope I have communicated the beauty in each human or generation coming up with their own takes on the same idea. It truly is special how different cultures and the evolution of the human race are intertwined by individuals' unique viewpoints, from coffee to the very physical laws that hold our universe together.





How Does Natural Research Develop?



Ashley

Did you know how prehistoric people could write scientific documents about nature? In today's ever-changing world, natural observation is a means of academic research within the historical flow, aimed at exploring Earth's operations.

How Did Ancient People Observe Nature?

Some prehistoric people observed nature and the night sky with their eyes, regarding the sun, moon, and stars. This helped them know when to plant crops and hunt animals.

1. Ancient Egyptian

Ancient Egyptians are known for their towering pyramids, magnificent temples, and powerful pharaohs. However, Egypt's extraordinary progress was one of the most remarkable aspects of this civilization. Specifically, Egyptian astronomers observed the sun, moon, planets, and stars to understand seasonal changes and organize important activities, leading to the development of calendars. Moreover, the annual appearance of the star Sirius enabled them to predict the flooding of the Nile River and begin the agricultural year.

Moreover, Egyptian astronomers observed from the terraces of temples and palaces to look at the horizon. For example, in the papyri of Abusir, many astronomers observed the stellar time from the temple terrace.

2. Ancient Greek

When the number of stars increased over time, the night sky was gradually mapped out with distinct constellations.

- Therefore, the Greeks used the first and last stars' locations, which enabled them to signal the appropriate time for daily activities (especially agriculture).
- Simultaneously, Greek societies used the moon for their civil and religious calendar and regular agricultural festivals.
- Agriculture is tied to the seasons, but the star cycle matches the sun, and the moon works differently.

The development of star calendars contained a large number of constellations. It enabled them to synchronise the divergent seasonal and lunar timetables.

Examples illustrate how particular stars might have been linked to particular divinities and festivals.

Observational Instruments and Implementation

Ancient people used to utilize simple instruments for natural observation.

According to the Instituto de Astrofísica de Canarias, José Lull (an ancient Egyptian astronomer) wrote about the observational instruments:

- Merkheth (the piece of wood) was placed horizontally, from which a plumb line hung to track stars across a level horizon and determine true north.
- Bay (a palm frond on the upper part) helped them make a Groove for the observational instrument.

Two objects were combined for observing to help priests-astronomers set accurate building orientations and measure time after sunset. It plays a crucial role in regulating temple rituals and state ceremonies. Moreover, it illustrated the existence of two other instruments:

- a type of gnomon, appearing drawn on many astronomical roofs
- a type of grommet could well correspond to the strange object crowning the head of the goddess Seshat.

The Egyptian astronomers also used clepsydras, as well as clocks using the Sun and shadow. Moreover, they contained lists of stars with information about risings, settings, and culminations.

How did it impact?

In the past, there were many innovations and information from nature observation, such as Sonar, Bullet Trains, Velcro, and Lotus Paint.

How Do People Observe Nature In Recent Times?

1. Tools

Natural Observations nowadays can be specialised in many ways: creating artificial natural environments, experimental laboratories, and others. As opposed to ancient natural observation, it brings about more observational techniques. There are two common natural instruments to observe nature:

- Flash Forever: a tool pioneering drone technology, aimed at reforesting inaccessible and dangerous locations, as well as delivering seed pods, data on reforestation success.
- Chloris Geospatial: a tool maintaining accuracy of carbon data to creating a high-integrity voluntary carbon market.

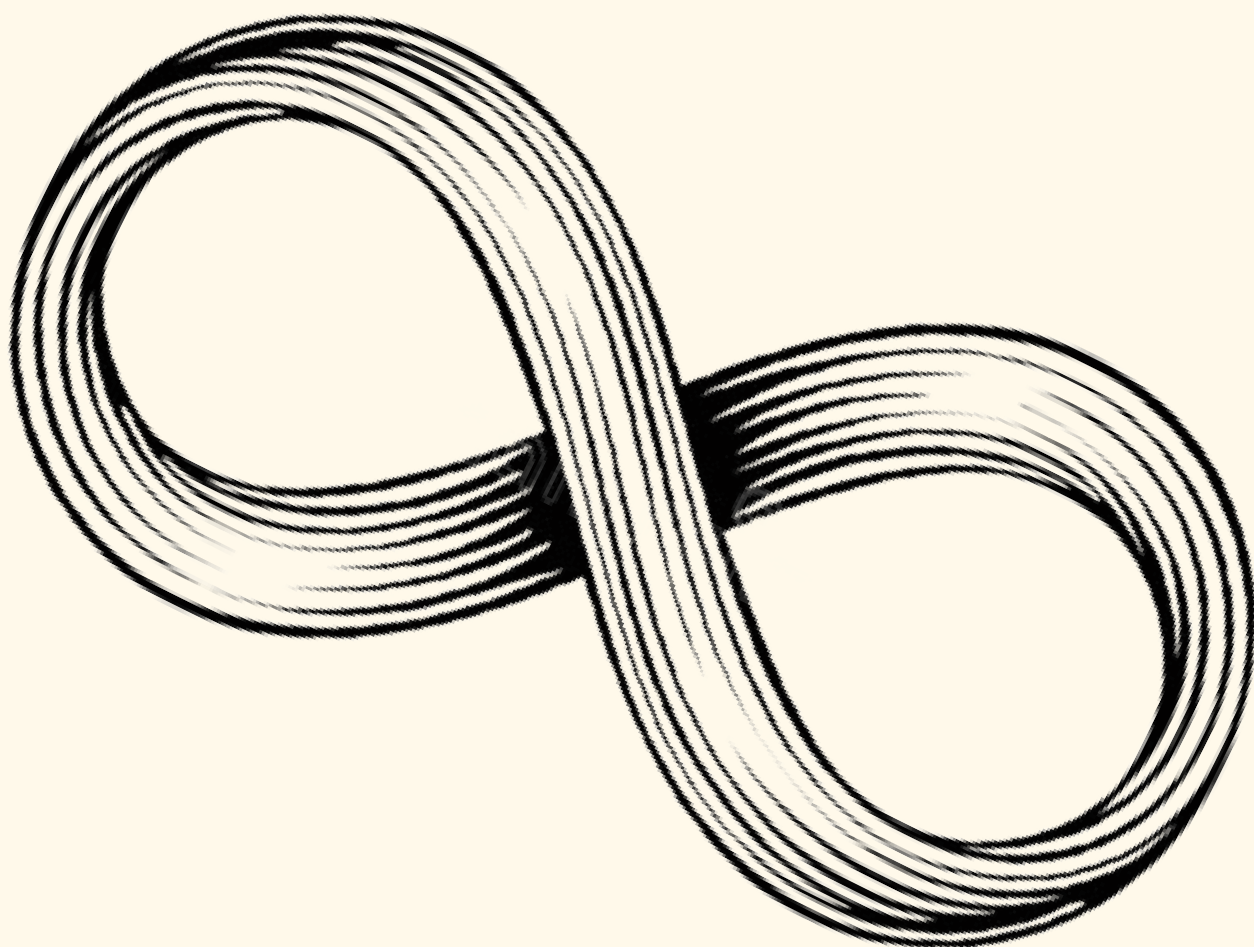
2. Significant Changes of Natural Observation

- Without interference or influence from the observer.
- Gain insights into behaviors without evidence in controlled settings and laboratories.
- Offers the advantage of capturing authentic behaviors with challenges.
- The presence of uncontrolled variables in natural settings can affect results.

Naturalistic observation remains unchanged variability to understand complex behaviors in real-world contexts.

Conclusion

In an ever-changing world today, natural observations can be exemplified by many challenges, tools and awareness and have undergone significant changes. However, it illustrates that humans always desire to broaden the Earth's cycle.



THE DUALITY OF ∞
WHERE INTUITION MEETS PROOF

Alisha Bilal

I would like you to imagine asking a group of children just starting primary school “What is the largest number you know?”. Of course, with the limited mathematical information available to a child at the age of 5, typical answers may have included 100, 200, maybe even a lucky 500 from a particularly ambitious responder. However, if you instead ask this same question to a group of 10 year olds, for example, the answer you are much more likely to hear is ‘infinity’. Yes, this is a word the vast majority of us become familiar with in our childhood, but do we actually understand infinity, and what does understanding infinity even entail? This article dives into the duality of mathematical intuition and proof, exploring when intuition can introduce us to the core of infinity, and when we are instead prompted to use precise mathematical reasoning to deepen our understanding of this iconic mathematical idea.

The building blocks of ∞

In order to truly understand infinity we should have a clear and simple definition of it to begin with. The Cambridge Dictionary states that infinity is “**unlimited space, time, or amount, or a number large beyond any limit**”. Essentially, infinity never ends, which is precisely why it is not simply an extremely large number, rather a concept which conveys a state of neverendingness. Hence, we instead use the symbol, ∞ . Understanding the concept behind this famous symbol to its fullest is likely the most digestible part of getting to know infinity; this is where we can use our intuition, or at least where infinity rationally follows our predisposed ideas of mathematics. Without much mathematical rigour, it is easy to see why the phrase “There are infinitely many natural numbers” is true. Counting 0, 1, 2, 3, 4... and so on will lead many of us to the conclusion that this list has a size of infinity, since it exceeds any known limit. In mathematics, rather than trying to process arbitrarily large numbers, we need to be more concise with our understanding of infinity. Therefore, using an area of mathematics called set theory, we can identify the set of natural numbers as an infinite set.

Comparing infinite sets

When infinity first starts breaking the rules, however, is when we begin to compare infinite sets with each other. A common method of comparison in set theory involves a process of matching elements in one set with another and checking for a bijection, also known as a one-to-one correspondence between the items in set A and set B. If such a bijection does exist, these sets are of the same size. In set theory, size is widely understood as the cardinality of a set, which describes the number of elements in a set, even when a set is infinite. The process of pairing up items in a list tends to be less algebraically rigorous compared to most modules encountered in high school maths - the skill it actually requires is a deeply imaginative thought process.

$\mathbb{N}$ vs $\mathbb{Z}$

A simple example we can start with to understand what a bijection looks like, and how it can subvert our expectations, is by comparing the set of all integers with the set of natural numbers.

N						0	1	2	3	4	...
Z	...	-4	-3	-2	-1	0	1	2	3	4	...

Figure 1a: A comparison between the natural numbers (row 1) and the integers (row 2). Note that there are two definitions of the set of natural numbers, this example includes 0, as this is typically used in set theory, the branch of mathematics we are covering.

Upon first instinct, if I ask which of these sets is the larger, I'm sure first instincts will lead some of you to deduce that the set of integers has a greater cardinality than the set of natural numbers. More specifically, many believe upon first glance that the set of integers is double the length of the set of natural numbers. By checking for a bijection between the sets, it easily becomes clear that this assumption is incorrect.

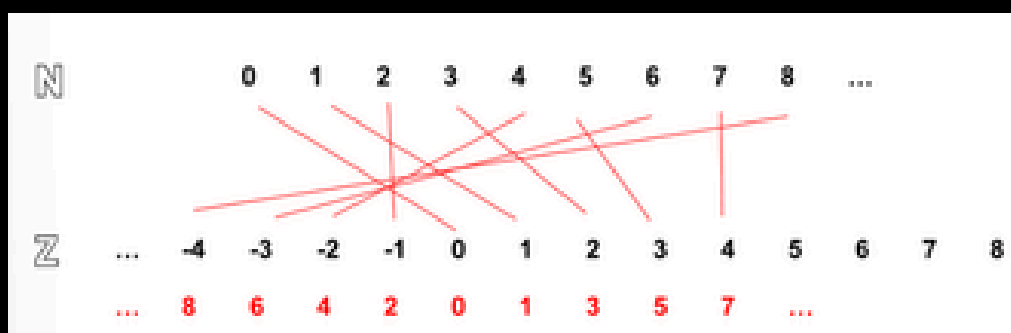


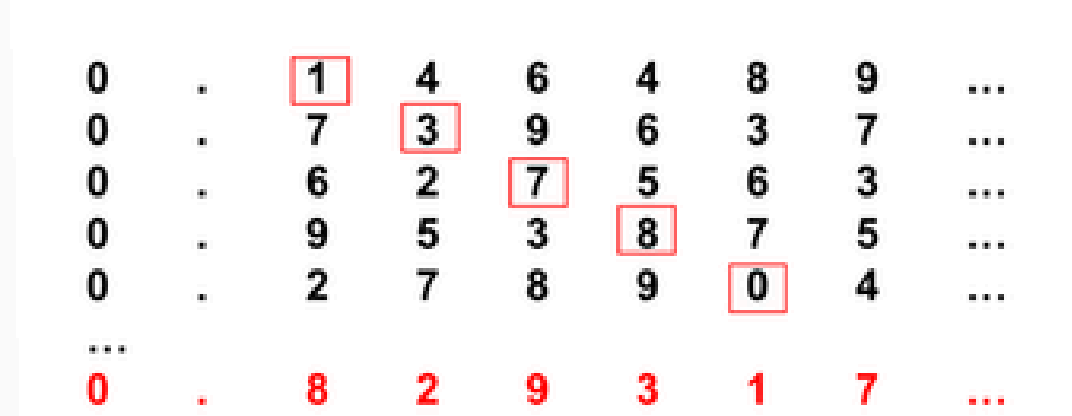
Figure 1b: A visual representation of the matching process of elements in the natural numbers with elements in the integers, with the red list below the integers representing the element in the natural number assigned to an element in the integers.

When actually pairing the natural numbers with the integers, it is best to arrange them in such a way that every odd number in the natural numbers is paired with a positive number in the integers, and every even number in the natural numbers is paired with a negative number in the integers. If we count both the set of even natural numbers and the set of odd natural numbers separately, it is clear that both of these are infinite sets. Therefore, the natural numbers can pair with the integers infinitely in both the positive and negative directions. As every natural number appears exactly once, and every integer also appears exactly once, we have constructed a bijection between the two sets, proving they have the same cardinality.

The surprising result we gain from this is that these two sets of numbers are indeed the same size, despite the integers containing both positive and negative numbers, while the natural numbers only include the non-negative numbers. In set theory, infinite lists such as the integers can be referred to as countable, which, by definition, is a set that has a one-to-one correspondence with the natural numbers. Practically, a countably infinite set can typically be identified if its elements can be arranged in a logical sequence, e.g. the prime numbers, as they can be arranged in a sequence in which every prime will eventually appear.

Can we count on $\mathbb{R}$?

In the case of a few other infinite sets, such as the set of all real numbers, difficulty arises when trying to create a comprehensive list. Even if we were to limit ourselves to just the real numbers between 0 and 1, it is impossible to form a sequence ensuring that every number is included. The set of real numbers is therefore our first example of what an uncountably infinite set looks like. Georg Cantor's diagonal argument proof provides a remarkably simple yet creative reasoning for why this is true.



A table illustrating Cantor's diagonal argument. The table consists of 6 rows and 8 columns. The first column contains the integer part (all 0s), and the second column contains the decimal point (all dots). The remaining columns contain digits. The first five rows represent existing numbers in a list, and the sixth row represents a newly constructed number. A diagonal line of red boxes highlights the digits from the first row's second column to the fifth row's sixth column. The digits in the sixth row are chosen to be different from the highlighted digits.

0	.	1	4	6	4	8	9	...
0	.	7	3	9	6	3	7	...
0	.	6	2	7	5	6	3	...
0	.	9	5	3	8	7	5	...
0	.	2	7	8	9	0	4	...
...								
0	.	8	2	9	3	1	7	...

Figure 2: Cantor's diagonal argument illustrated using a table of decimals. So long as each digit in a newly generated decimal differs from those in the red boxes (which follow a diagonal pattern) we can guarantee that the newly generated decimal differs from every number on the list so far.

As seen in the figure, by forming a table of decimals, we can visualise a diagonal line such that for the first decimal, the first digit is highlighted, and for the second decimal, the second digit, and so on and so forth. From here, if we want to create a decimal that is unique to the table, we can do so by using a different digit than the one highlighted in the diagonal line. By doing so, the new number differs from the first number in its first digit, the second number in its second digit, the third number in its third digit, and so on to the n th number in its n th digit. For this reason, this number will never be identical to any number on the original list. This proof and the subsequent discovery of uncountably infinite sets has led to one of the most counterintuitive conclusions within mathematics: Some infinities are larger than others.

At first glance, this statement can sound quite counterintuitive, especially as we previously debunked the common assumption that the set of integers is twice the size of the set of natural numbers. However, this concept ties directly back to cardinality, which truly comes into play once we consider uncountably infinite sets, and their differences from infinite sets which are countable. Using Cantor's diagonal argument, it is clear that no matter any effort to list every real number, a new one which is not already included in the list can be created. Thus, there is no one-to-one correspondence between the real numbers and the natural numbers, and as the real numbers cannot be listed with the natural numbers as indexes, the cardinality of the set of real numbers is strictly greater than that of the natural numbers.

This fascinating conclusion demonstrates how mathematical intuition and rigorous proof can work in harmony; an initially counterintuitive idea can be explored through intuition and further become established through clear reasoning. In reflecting upon the story of infinity, and how set theory helps us understand it, it is particularly interesting to notice the duality of intuition and rigorous proof, and how they intertwine within the discoveries regarding infinity. Perhaps a considerable takeaway from understanding infinity is the importance of testing our intuition. In the initial stages, intuition may guide us towards the mathematical truth, as seen in Cantor's proof. However, rigorous mathematical proof determines whether an idea can survive intellectual scrutiny. From understanding infinity, knowing when to trust intuition alone is useful – but knowing how and when to question it is essential.



From Human Expertise to AI Algorithms

How Generations of Medical Knowledge Are
Shaping the Future of Diagnosis



Annie

When you walk into a hospital today to have a symptom diagnosed, the chances are you will meet with your physician, get examinations done, and receive your diagnosis within a few weeks. But what you may not see is the technology utilized in the process—how medical advancements in AI are changing the future of healthcare altogether.

History of diagnosis

In the early 1900s, medical diagnosis heavily relied on physical examinations and observation from doctors. Symptoms were handwritten in paper medical records, and basic tools such as the stethoscope, thermometer, and lab tests were the only tools available to assist medical examinations. As a result, diagnosis was often inaccurate and likely to align with societal biases during the time, discriminating against patients of different races and genders.

However, technological development rapidly approached the Gen X in the 1990s. Compared to the traditional methods of observation, sophisticated imaging technologies such as X-rays, CTs and MRIs paved the way for evidence-based diagnosis. These tools assisted physicians in making crucial medical decisions. Along with the advancements of medical tools, early paper medical records became digitized, retrieving and storing historical records in an electronic system. This helped doctors compare and interpret data from past diagnoses.



Utilizing AI in medical diagnosis

In contrast to Gen X, current technological advancements utilize AI to deeply analyze patterns and make predictions based on electronic records. Specific examples of AI technology in the medical field include:

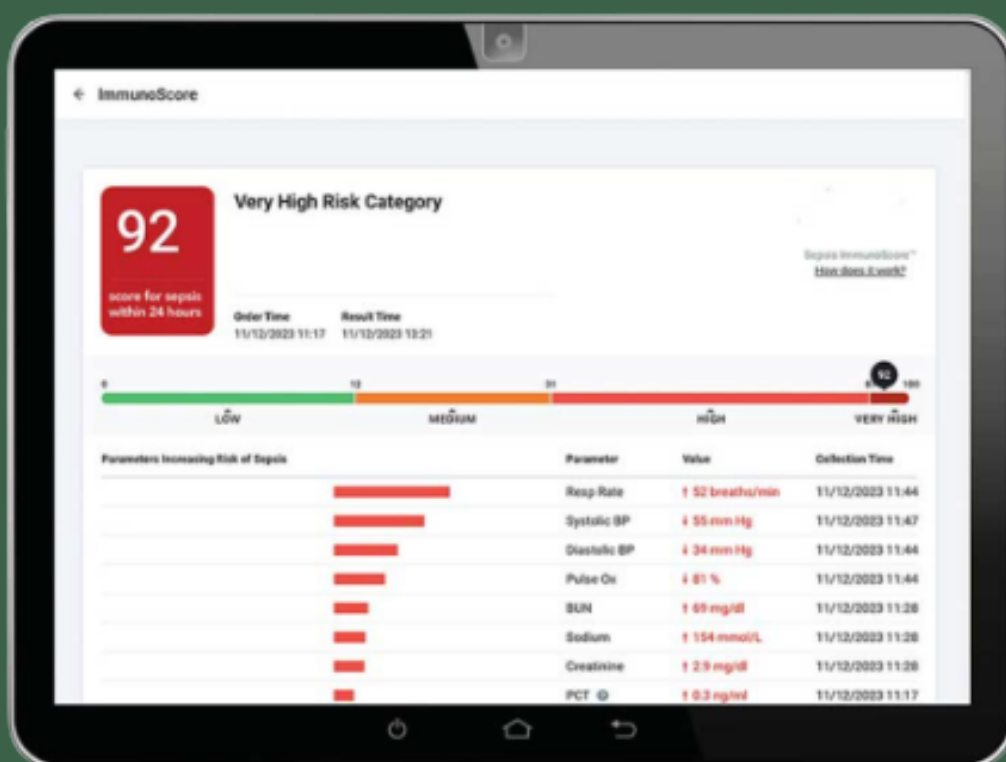
- Aiodic in Medical Imaging

Provides radiologists with real-time assistance. Automatically analyzes scans like CTs, X-rays, and MRI's to flag urgent conditions, coordinates team alerts, and triages critical cases.

- Sepsis Immunoscore in early risk tracking

The first FDA-authorized AI-driven tool designed for early sepsis risk tracking. It uses electronic medical records and inspects 22 parameters to predict a patient's risk of developing sepsis within 24 hours.

These AI technologies allow for accurate and efficient predictions in medical monitoring; however, AI does not yet completely replace human expertise because technology is still developing, and ethical considerations require human judgement when making decisions that directly affect patients.



Predicting the Future of Diagnosis

As AI continues to develop, medical diagnosis is likely to become increasingly accessible and personalized. Some new technologies currently being worked on or that can be anticipated in the future are personalized medicine tailored to individual genetic profiles, real-time monitoring that allows healthcare providers to continuously check for health risks, and remote online diagnosis that can be focused on expanding healthcare access to underserved or geographically isolated communities.

However, the increasing use of AI in medicine also raises important ethical concerns. First, early medical records can introduce a potential bias in AI algorithms. AI systems learn from existing medical data, meaning that the biases in race or gender reflected across previous historical records could potentially be amplified through AI predictions. Second, the aspect of accountability. Who will be held accountable when an AI makes a mistake? Should society allow a machine to decide your medical treatment? Lastly, there are rising concerns about the risks to employment in an AI-dominated field. As AI becomes capable of performing multiple and extensive tasks, many responsibilities traditionally performed by healthcare professionals may become replaced.

As society continually progresses, technology should be developed in a way that works alongside and empowers healthcare rather than replacing or dominating human expertise. Maintaining human oversight is crucial to ensuring that new medical technologies remain both effective and ethically responsible for the future of healthcare.





Sanger Sequencing vs. Next-Generation Sequencing

How DNA Sequencing Has Transformed
Modern Biology and Medicine

Esther Tyrnauer



Introduction

In 2003, an international team of scientists completed a high-quality reference sequence of the human genome after 13 years of work. The Human Genome Project was far more than a single sequencing run: it involved genome mapping, sequencing, development of new technologies, studies of model organisms, and research into the ethical, legal, and social implications of genomics. The project ultimately cost about \$2.7 billion in U.S. funding. Today, by contrast, a laboratory can generate a whole-genome sequence in a much shorter time and at a tiny fraction of the cost of the overall Human Genome Project. The comparison therefore illustrates not that one genome once required thirteen years to sequence, but how dramatically sequencing technology, automation, and computational analysis have changed since the project began. It happened because DNA sequencing itself was fundamentally transformed.

Every cell in the human body carries a set of instructions written in a four-letter chemical code: A, T, C, and G. For most of human history, that code was unreadable; scientists could observe the traits it produced but not the molecular language behind them. Frederick Sanger changed that in 1977, when he developed the first reliable method for reading a DNA sequence base by base. Three decades later, a second revolution arrived: next-generation sequencing (NGS), a family of technologies capable of reading millions of DNA fragments at once. Where Sanger sequencing made genomics possible, next-generation sequencing made it practical – and that shift, from a rare, expensive laboratory feat to a routine clinical and research tool, has reshaped how cancer is treated, how rare diseases are diagnosed, and how the world tracks infectious diseases. This essay argues that the leap from Sanger sequencing to next-generation sequencing was not merely a matter of speed or cost, but a fundamental shift in scale that turned DNA sequencing from a specialized scientific milestone into one of the most transformative tools in modern medicine.

The Development of Sanger Sequencing

Sanger sequencing, developed by Frederick Sanger and his colleagues in 1977, was the first widely successful method for determining the precise sequence of nucleotides in DNA. The technique relies on a process called chain termination. A DNA sample is combined with DNA polymerase, primers, normal nucleotides, and chemically modified nucleotides called dideoxynucleotides (ddNTPs). As the polymerase builds new DNA strands, it occasionally incorporates a ddNTP instead of a normal nucleotide. Because ddNTPs lack the chemical group needed to attach the next base, the strand's growth stops at that point. This produces a mixture of DNA fragments of different lengths, each ending in a particular base. By separating these fragments according to size and determining their terminal bases, scientists can reconstruct the DNA sequence. The method was foundational to modern genomics and was ultimately used, with substantial technological improvements, during the Human Genome Project.

Sanger sequencing's historical impact is hard to overstate. The Human Genome Project used Sanger-based sequencing alongside extensive mapping and other technological advances to construct a human reference sequence. However, sequencing large genomes with this approach required processing enormous numbers of individual DNA fragments and assembling them into a complete sequence. These constraints – precision without the massive parallelism of newer technologies – helped create the demand for a fundamentally different approach.

The Rise of Next-Generation Sequencing

Next-generation sequencing fundamentally reimagined how sequencing could be done. Rather than processing DNA fragments sequentially, short-read NGS platforms sequence enormous numbers of fragments in parallel through a process called massively parallel sequencing. Illumina's sequencing-by-synthesis approach, for example, uses DNA fragments attached to a flow cell, bridge amplification to create clusters, and fluorescent reversible-terminator nucleotides to determine the sequence as bases are incorporated. A landmark 2008 study demonstrated that this massively parallel approach could generate billions of bases in a single experiment and apply the technology to whole-human-genome sequencing.

The differences between the Human Genome Project and modern sequencing need to be framed carefully. The Human Genome Project took about 13 years and cost approximately \$2.7 billion in total U.S. funding, but that figure covered a broad scientific program rather than a single sequencing run. Modern sequencing instruments can generate enormous quantities of sequence data in hours or days, while the cost of sequencing an individual genome has fallen dramatically. NHGRI describes this broader technological progress as involving roughly a million-fold reduction in the cost of sequencing a human genome since the beginning of the field. NGS also offers far greater scalability, allowing researchers to sequence entire genomes, exomes, or targeted gene panels for many samples in a single run rather than one gene or fragment at a time. This combination of speed, affordability, and throughput transformed sequencing from a specialized, resource-intensive endeavor into a routine laboratory and clinical tool.

Transforming Biology and Medicine

The scale and affordability of NGS have driven transformative changes across biology and medicine. In cancer research, NGS allows scientists to sequence a tumor's DNA and compare it to a patient's normal tissue, revealing mutations associated with an individual's cancer. This has enabled the identification of actionable mutations, such as alterations in EGFR or BRCA-related genes, that can help guide targeted therapies rather than relying entirely on one-size-fits-all treatment.

In rare disease diagnosis, NGS has been especially powerful. Many rare genetic disorders previously went undiagnosed for years because testing one gene at a time was impractical. Whole-exome and whole-genome sequencing now allow clinicians to examine a patient's protein-coding regions or entire genome at once, helping shorten the diagnostic process for many families.

NGS has also become central to precision medicine, an approach that uses genetic information to help tailor treatment decisions to individual patients. In infectious disease surveillance, NGS proved essential during the COVID-19 pandemic: public health agencies such as the CDC used whole-genome sequencing of viral samples to identify SARS-CoV-2 variants and track their spread, informing public health responses. Beyond medicine, NGS has expanded genomic research broadly, enabling large-scale studies of population genetics, microbiomes, and evolutionary biology that would have been difficult or impossible with earlier sequencing approaches.

The Future of DNA Sequencing

DNA sequencing continues to evolve beyond conventional short-read NGS. Long-read sequencing technologies, including those developed by Pacific Biosciences and Oxford Nanopore, can read DNA fragments thousands to hundreds of thousands of bases long. Because of their distinctive read lengths and single-molecule approaches, long-read platforms are commonly classified as a distinct third-generation sequencing (TGS) approach rather than simply being treated as another form of standard short-read NGS. Their long reads are particularly useful for resolving repetitive and structurally complex regions of the genome that short-read sequencing can struggle to assemble.

This distinction also helps explain why long-read sequencing represents more than simply a faster version of NGS. Short-read sequencing revolutionized genomics through massive parallelization, while third-generation long-read methods introduced different strategies for reading DNA molecules and addressing limitations caused by short reads. These technologies have contributed to the completion of previously difficult regions of the human genome, including repetitive regions that could not be fully resolved during the original Human Genome Project.

Artificial intelligence is increasingly woven into genomic analysis, helping researchers interpret the flood of data that sequencing now produces. Machine learning models can help predict the functional impact of genetic variants, identify potentially disease-associated mutations, and accelerate research into drug targets. Combined with falling sequencing costs, these tools point toward a future where personalized medicine – treatment plans informed by an individual's genetic characteristics – becomes increasingly common.

Still, significant challenges remain. Sequencing technology is not equally accessible worldwide, and disparities in access to genomic testing risk widening existing health inequities. The sheer volume of genomic data being generated also outpaces our ability to interpret it fully; many genetic variants remain of uncertain significance because their biological effects are not yet understood. Finally, the growing use of genomic data raises pressing ethical questions around privacy, consent, and potential genetic discrimination, all of which society will need to address as sequencing becomes further embedded in healthcare.

Conclusion

A technology that once required a thirteen-year, international research program to produce a human reference sequence can now generate an individual genome sequence in a fraction of that time and at dramatically lower cost. But the comparison is not simply between “thirteen years” and “one day”: the Human Genome Project was a vast scientific undertaking involving much more than sequencing alone. Sanger sequencing established the ability to read DNA accurately; short-read NGS transformed the field through massive parallelization; and third-generation long-read sequencing is now addressing genomic regions that earlier technologies struggled to resolve. Together, these advances have turned genome sequencing from an extraordinary international project into a routine tool of modern biology and medicine. But cheaper and faster is not the same as fairer or wiser: a technology this powerful only delivers on its promise if access, interpretation, and ethical safeguards keep pace with the science. What we choose to do with the ability to read the genome so rapidly is the question this generation of scientists – and this reader, perhaps – will help answer.



The Study of Sleep Tracking Technology

How Has Technology Changed the Way We Understand Sleep?



Dane Baes

Introduction

Sleep is the natural state of rest that practically every creature on Earth needs and makes up a good margin of your life. Therefore, it is something that has to be cared for and maintained well through a good sleep schedule and proper routines. But to expand on that care, individuals have developed methods of technology that have helped us advance our sleep experience in ways that we didn't know. This can, in turn, improve our quality of life when one gets enough rest, showing how essential getting a good night's sleep can be.

For most of human history, waking up depended on natural light, household routines, or simple mechanical alarms. Today, a smartwatch can record movement, estimate sleep stages, and turn a night of sleep into graphs and numbers. Sleep science has changed from simply asking, "When did I wake up?" to asking, "What happened inside my body while I was asleep?" sleep is more than the number of hours spent in bed. Scientists study sleep timing, stages, movement, heart rate, breathing, and brain activity to understand how the body and brain recover. Modern sleep trackers make some of this information available outside a laboratory, but their measurements are not identical to clinical sleep studies. Technology has evolved from simply waking people up to collecting detailed information about sleep, but the scientific goal remains understanding how sleep affects the body and brain.

From Alarm Clocks to Measuring Sleep

To begin, traditional alarm clocks were mainly designed to control when people woke up, but they did not provide information about what happened during the night. People could know when they went to bed and when they woke up, but they had little information about the different changes happening while they were asleep. As sleep science developed, researchers began using more detailed methods to understand what happens during sleep. One example is polysomnography (PSG), which records brain activity along with other signals from the body to identify sleep and wake periods. Smith et al. explain that polysomnography is an important clinical method for evaluating sleep because it provides much more detailed information than simply observing whether someone appears to be asleep.

This development created a major shift in how scientists understand sleep. Instead of treating sleep as one continuous block of time, researchers began examining the different changes that occur throughout the night. Sleep consists of non-REM and REM stages, which have different patterns of brain and body activity. This helped scientists understand that sleep is an active process rather than simply a period when the body and mind shut down to rest. Before these advances in sleep science, people generally viewed sleep as a simple period of rest and had much less understanding of what was happening internally. Modern methods such as polysomnography have allowed researchers to study these changes in much greater detail (Smith et al.)

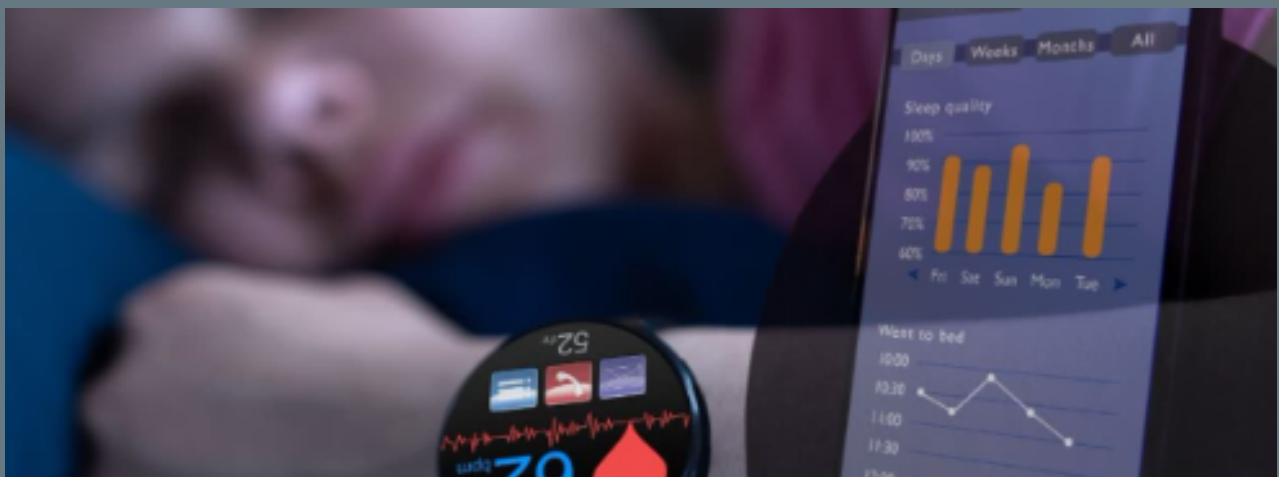


The Rise of Sleep Trackers

The modern state of wearable technology has changed how people measure their sleep by making it possible to collect sleep information from home rather than only in a sleep laboratory. Because devices such as smartwatches, fitness bands, and smart rings use sensors like accelerometers, heart rate monitors, and other physiological measurements to estimate when a person is asleep and identify general sleep patterns. According to Smith et al., actigraphy provides an objective way to monitor sleep over extended periods in a person's normal environment, making it a practical tool for observing everyday sleep habits rather than relying only on clinical testing.

Subsequently, an additional advantage of wearable technology is that it allows people to review their sleep data over time instead of relying only on their memory. Rather than trying to remember how well they slept, users can compare information from multiple nights to identify trends in their sleep schedule. As Yuan et al. explain, while actigraphy has limitations in identifying specific sleep stages, it is still useful for tracking overall sleep patterns across extended periods.

A real-world example of this can be seen through a high school student monitoring their sleep schedule. In the past, a student might have only used an alarm clock to wake up for school. Today, that same student could wear a smartwatch every night and review several weeks of sleep data to notice patterns in when they go to bed, wake up, or get less sleep before busy school days. Although the exact sleep-stage estimates should not be treated as a medical diagnosis, the device can still provide valuable insight into overall sleep habits and encourage healthier sleep routines (Smith et al.; Yuan et al.).



How Accurate Are Sleep Trackers, And Why Does Sleep Data Matter?

Scientists now understand that sleep is connected to much more than simply feeling rested. Research has connected sleep with memory, learning, attention, physical recovery, and daily functioning. This is especially important for teenagers because the CDC found that 72.7% of high school students in the United States were getting less than eight hours of sleep on school nights, even though teenagers are recommended to get 8–10 hours of sleep. Insufficient sleep has also been associated with problems involving attention, behavior, and academic performance (Centers for Disease Control and Prevention).

Wearable sleep trackers generally estimate sleep using movement and other sensor data. This can make them useful for identifying broad sleep patterns, but they have more difficulty determining exactly which sleep stage someone is in. Smith et al. explain that actigraphy can be useful for measuring sleep and wake patterns, but it has limitations compared with more comprehensive sleep measurements. More recently, Yuan et al. reviewed studies comparing wrist-worn actigraphy with polysomnography and found that actigraphy had some ability to classify different sleep stages, but the studies varied considerably in how they grouped sleep stages and measured accuracy.

This shows that modern technology can give people more information about their sleep, but having more information does not necessarily mean that the information is perfect. For example, a sleep tracker might show a pattern of shorter sleep over several weeks, which could help someone recognize their general sleep habits. However, its estimated amounts of REM, deep, or light sleep should be treated as estimates rather than a medical diagnosis. The research on actigraphy shows that these devices can be useful for observing general sleep patterns while still having limitations when measuring specific aspects of sleep.

Conclusion

The journey from the alarm clock to the sleep tracker shows how science has moved from controlling sleep schedules to measuring sleep in increasingly detailed ways. Mechanical alarms answered when it was time to wake up; modern wearables attempt to answer what happened during the night. However, newer technology does not automatically mean perfect measurements. Research shows that wearables can be useful for tracking broad sleep patterns, but they can differ from laboratory methods, especially when estimating individual sleep stages. The future of sleep science will likely depend on combining convenient wearable technology with rigorous scientific validation. Ultimately, the biggest change is not the alarm clock itself—it is the amount of information people can now collect about something that was once difficult to observe outside a laboratory.

Marlee Mei



The Evolution of Medical Records

Transitioning from paper files to digital databases



Introduction

In today's world of technology, media, and AI, many healthcare workers are shifting from traditional methods of paper-keeping, charting, and even direct care to digital platforms and websites, including telehealth and prescribing tools. These innovations can streamline information and enhance efficiency, as well as benefit hybrid approaches.



Electronic Medical Records (EMRs)

Firstly, electronic medical records (EMRs) reduce the need for paper files, which can get lost, damaged, or burnt. They offer a secure workflow when paired with strong, private internet connections—though this can raise privacy and security concerns that must be addressed. Healthcare providers should be transparent with patients about how data is being stored and ensure compliance with regulations. EMRs can be easily searched and organized into digital folders, allowing for easier grouping, tracking, and identification of duplicates, since records are constantly being updated. Moreover, data can be timestamped, which creates a clear history of patient care. Beyond organization, EMRs reduce storage, unnecessary paper, and administrative work. Digital signatures streamline authorization processes, while medical coding improves accuracy and consistency across multiple providers. Additionally, AI can assist with charting and diagnosing, such as analyzing images, which is far more difficult to do on paper records.



Paper Charting

However, despite technological advancements, many providers still rely on traditional paper charting, which offers many advantages. One clear advantage of paper charting is its authenticity and reliability. Paper files are often considered more legitimate since they contain real signatures, stamps, and seals, which are hard to forge. Additionally, paper records can be stored in separate, secure paper file rooms, protecting them from cyber threats that digital systems are vulnerable to. Digital content can become accidentally deleted, corrupted, or disorganized without proper management and training. Yet, paper charting is straightforward and doesn't require much technical expertise. It is efficient for quick notes during patient visits, making it essential for backup in the event of an internet outage or system failure. This ensures that technical issues never compromise patient care.

EHR-Related Patient Safety: Key Numbers

48%

Fewer medication errors
post-EHR

44%

Serious med errors cut with
CPOE

3.6%

Dosage alert decrease via
CPOE

40-96%

Clinical alert override rate
(est.)

Hybrid

The hybrid approach is becoming popular as health networks find solutions to balance the two methods while keeping things efficient. Many clinics maintain paper records as a reliable backup while using digital systems as their main ecosystem for administrative tasks. This approach allows for specialized care, and an example is that some Traditional Chinese medicine doctors continue to use handwritten notes for observation and prescription while using digital platforms for scheduling and data exchange among other providers. However, paper charting is not perfect. A significant concern is the risk of illegible handwriting, particularly on prescriptions. If a medical doctor's handwriting is difficult to read, it can cause errors and misunderstandings. This would also extend the time for the medication to reach the patient, as there would be extra communication and clarification between doctors and pharmacists. For this reason, many clinics and hospitals are moving toward digital prescribing tools. While these tools may be less common and familiar to some patients and providers, the benefits are clear. Not only are prescriptions instantly written, but they can be sent directly to pharmacies, speeding up refills and getting medication to patients.

Conclusion

Health networks should incorporate both traditional and modern practices. Keep paper for backup and quick notes on the go, and use digital systems to oversee everything and assist with analysis. Picking a system that can exchange data with other providers is essential; for some, sending paper files can be bothersome or delayed. In all, transparency is important, and patients should know how their information is stored.



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“Duality is an essential feature of nature;
we cannot have light without darkness,
joy without sorrow, or life without death.”

— *Alan Watts*